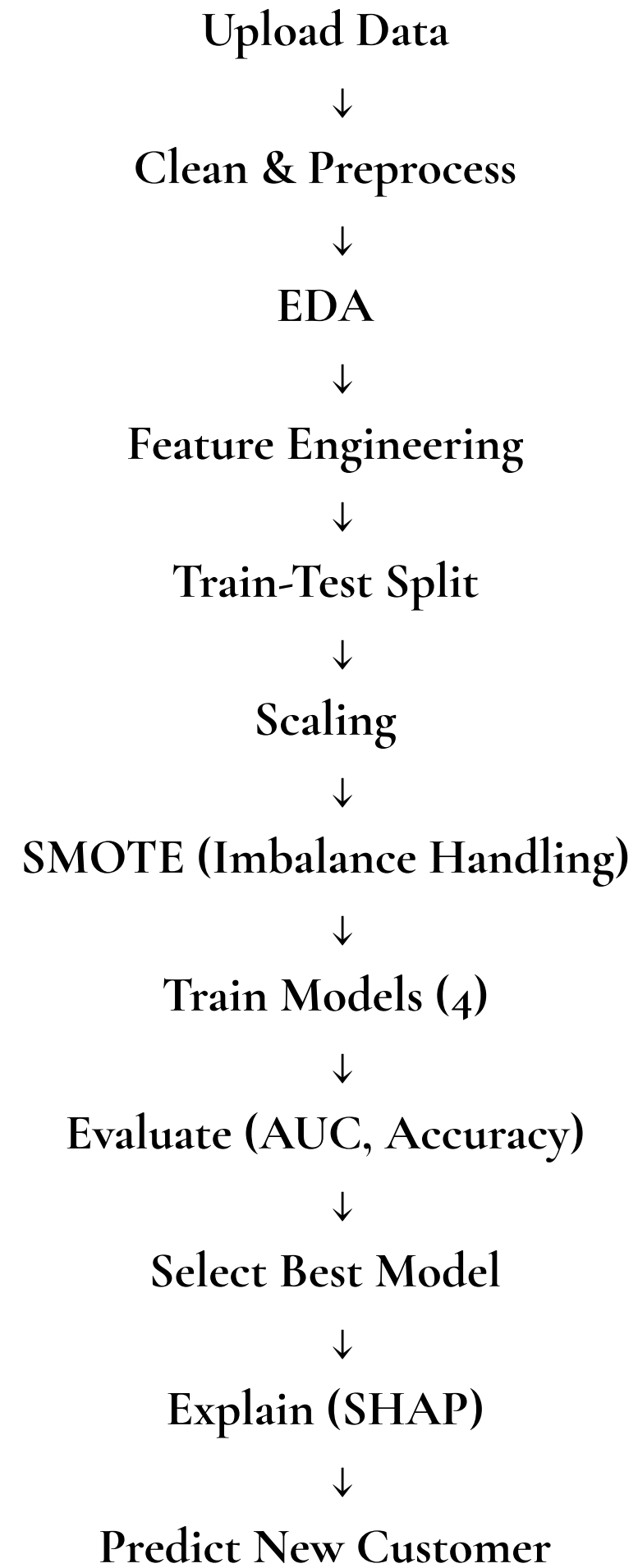


Customer Churn Prediction for Telecom Companies

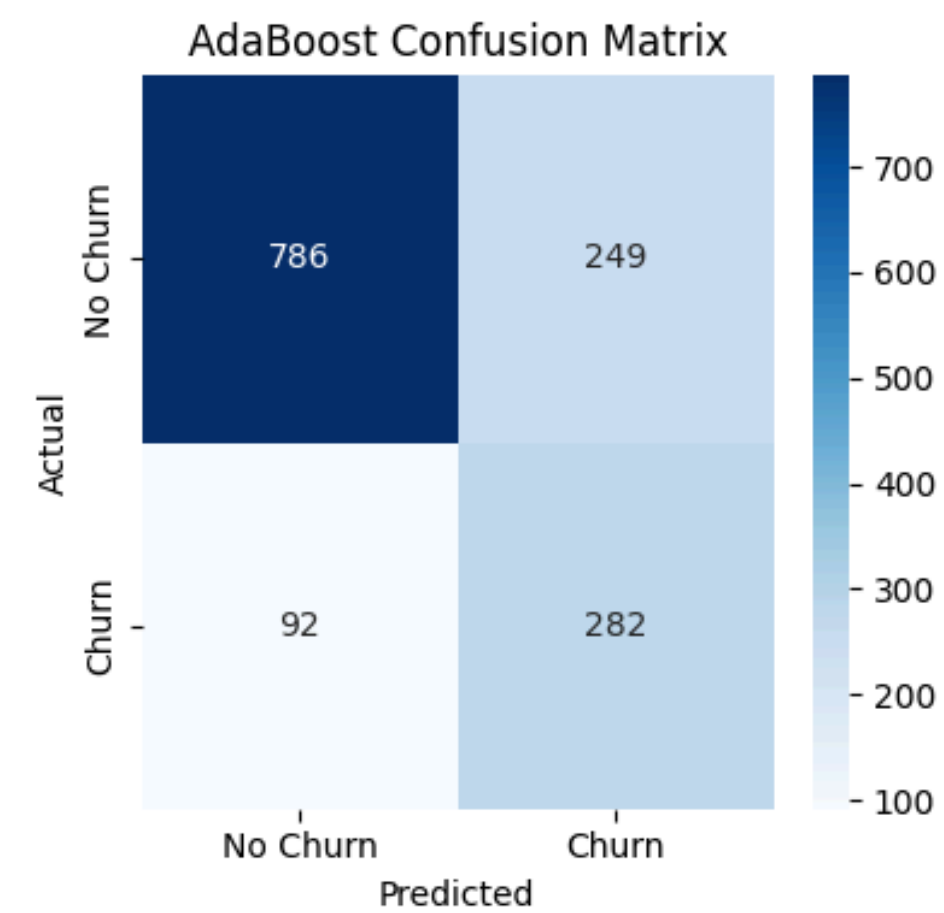
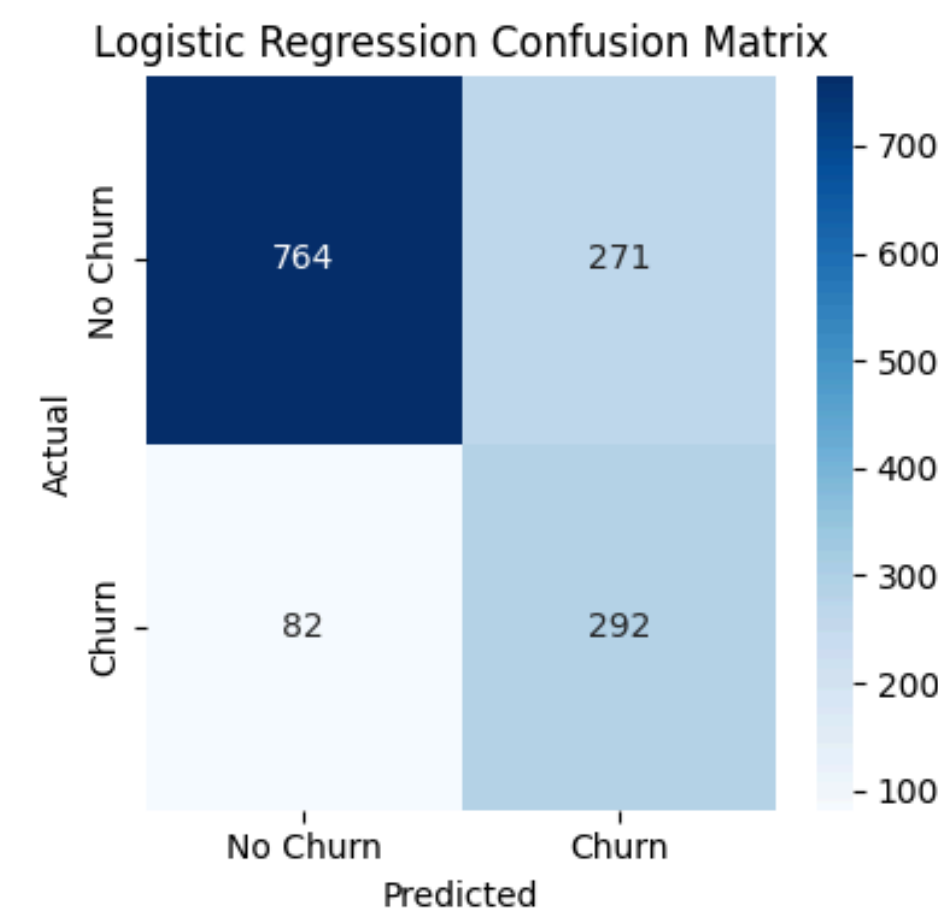
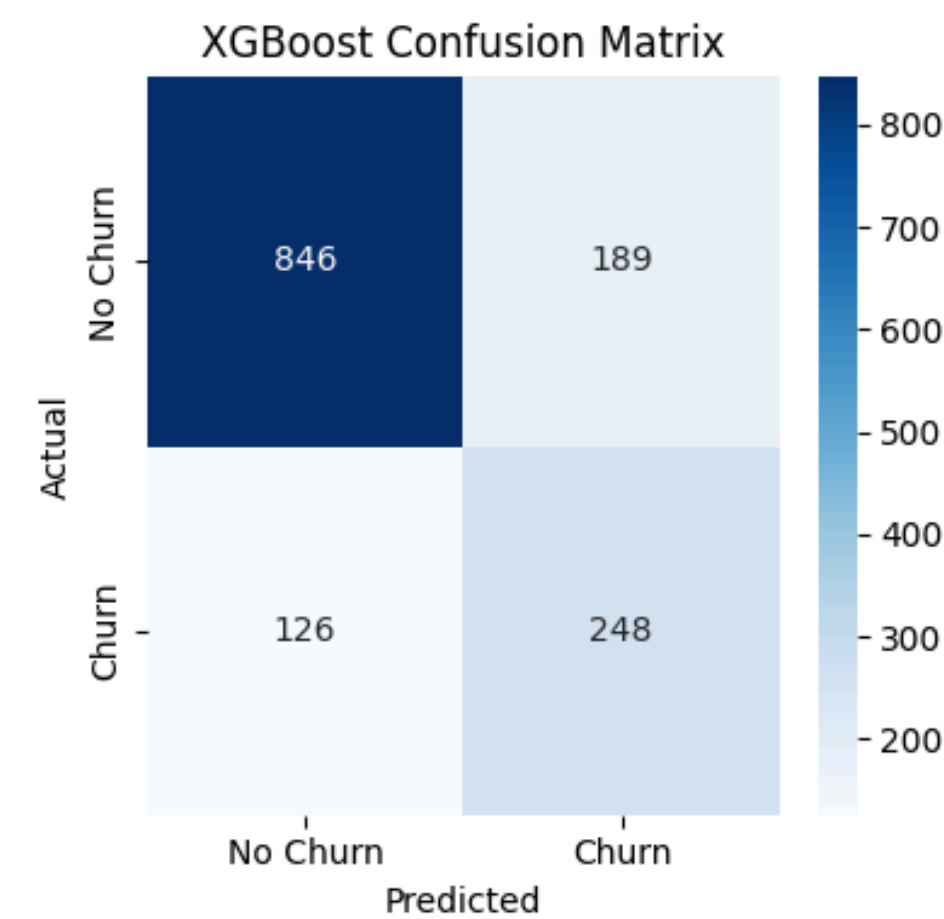
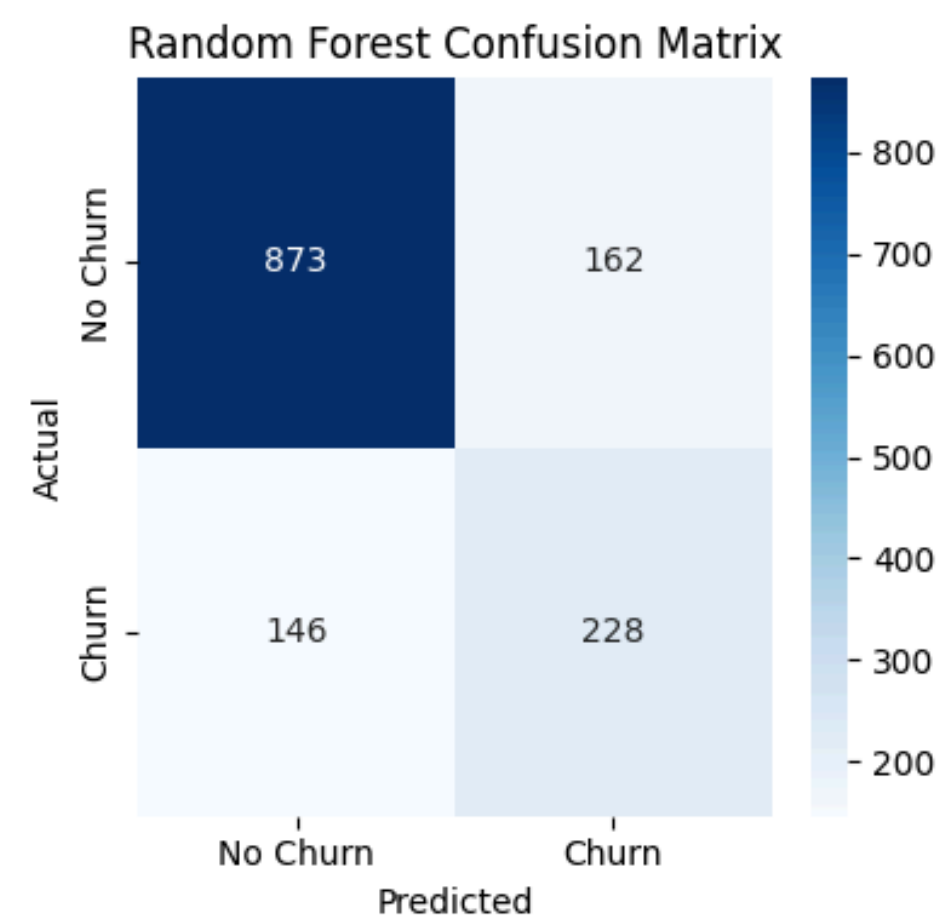
**CAPSTONE PROJECT 1: BUILDING A MACHINE LEARNING MODEL TO
PREDICT TELECOM CUSTOMER CHURN AND HELP COMPANIES RETAIN
SUBSCRIBERS.**

Arun PS
Uzma Haseeb
Rehan Biju

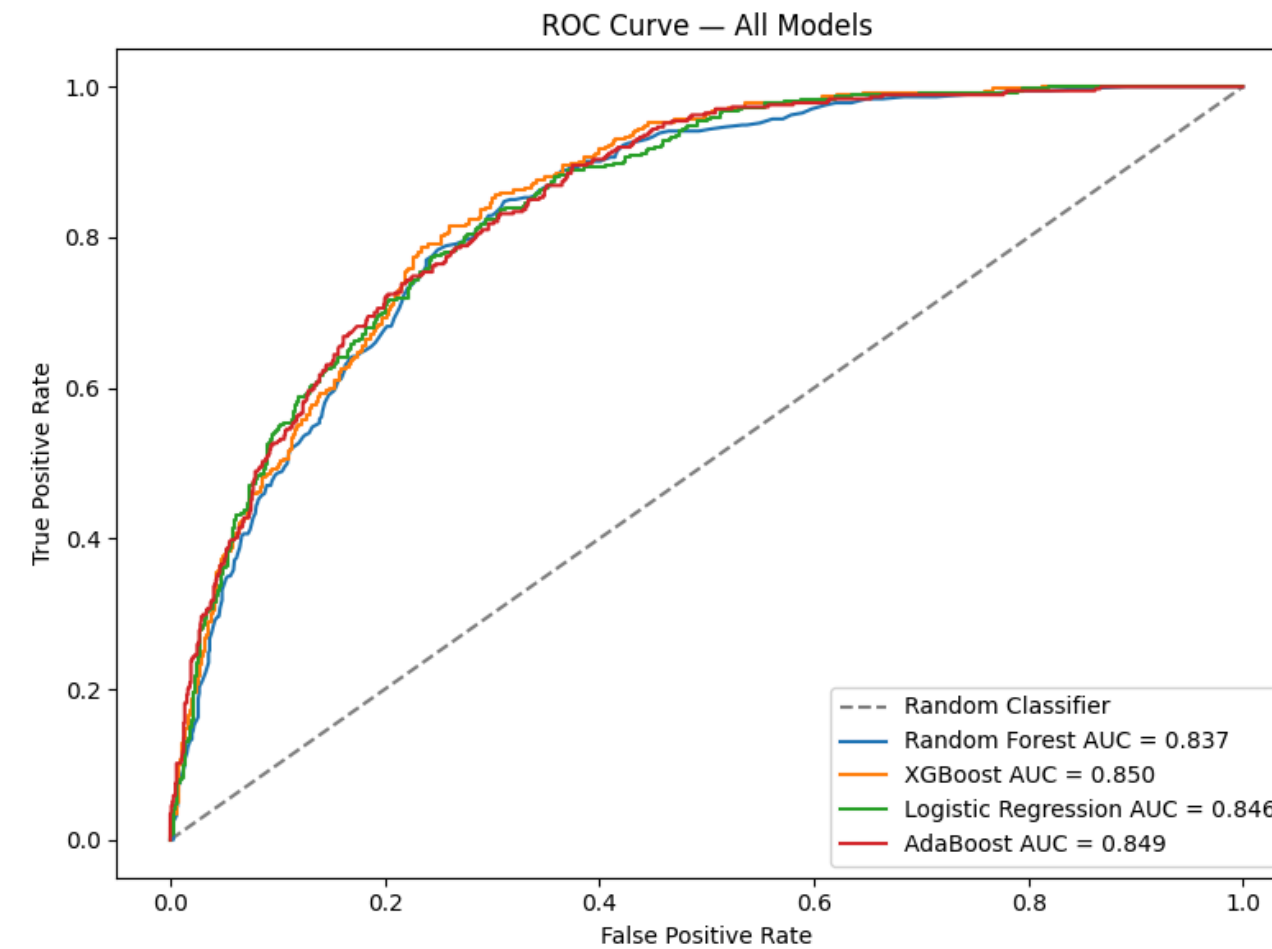
Methodology



Model Performance

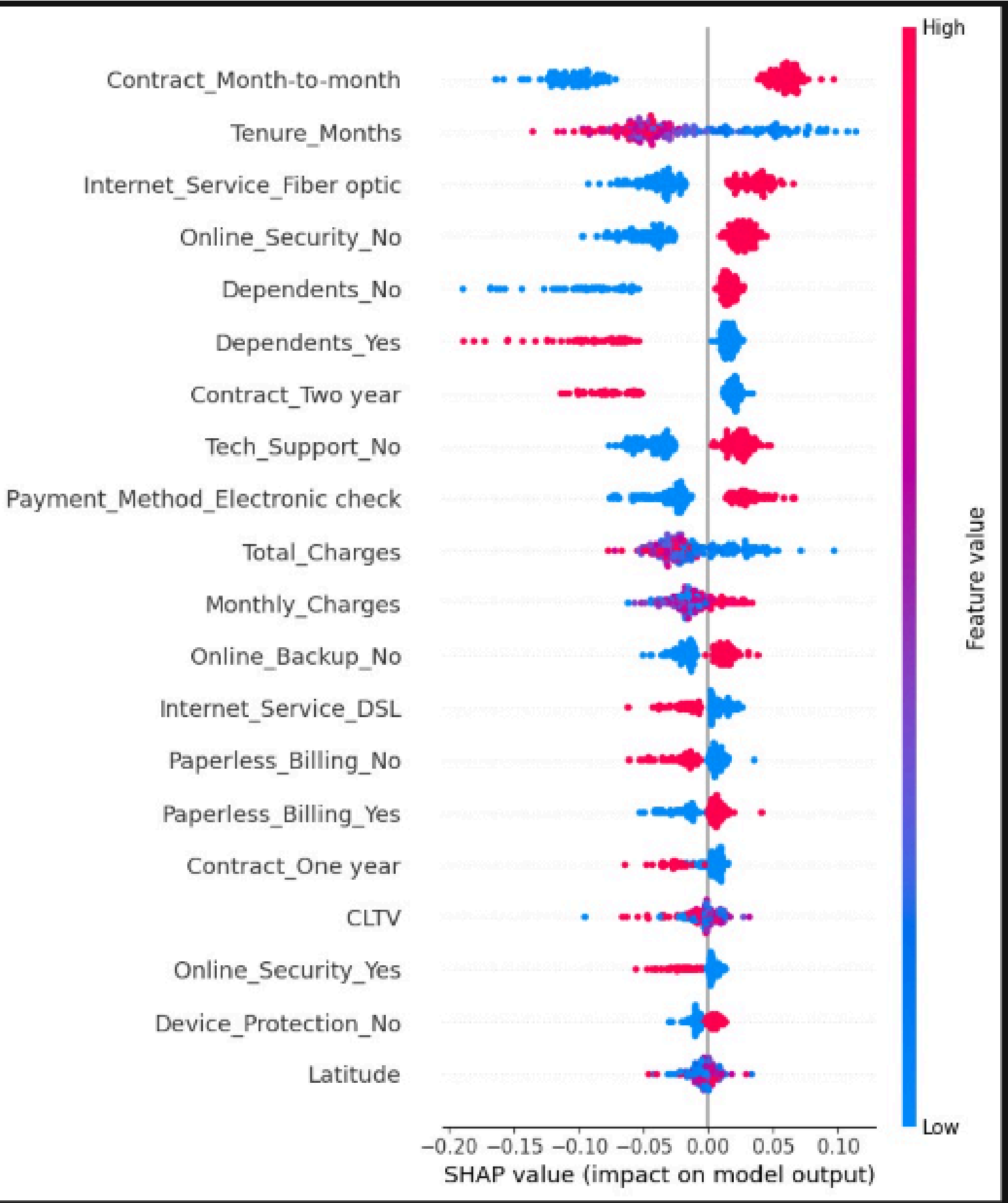


Evaluation



Model	Accuracy	Precision	Recall	F1-Score
Random Forest	~78%	~58%	~61%	~60%
XGBoost	~78%	~57%	~66%	~61%
Logistic Regression	~75%	~52%	~78%	~62%
AdaBoost Classifier	~76%	~53%	~75%	~62%

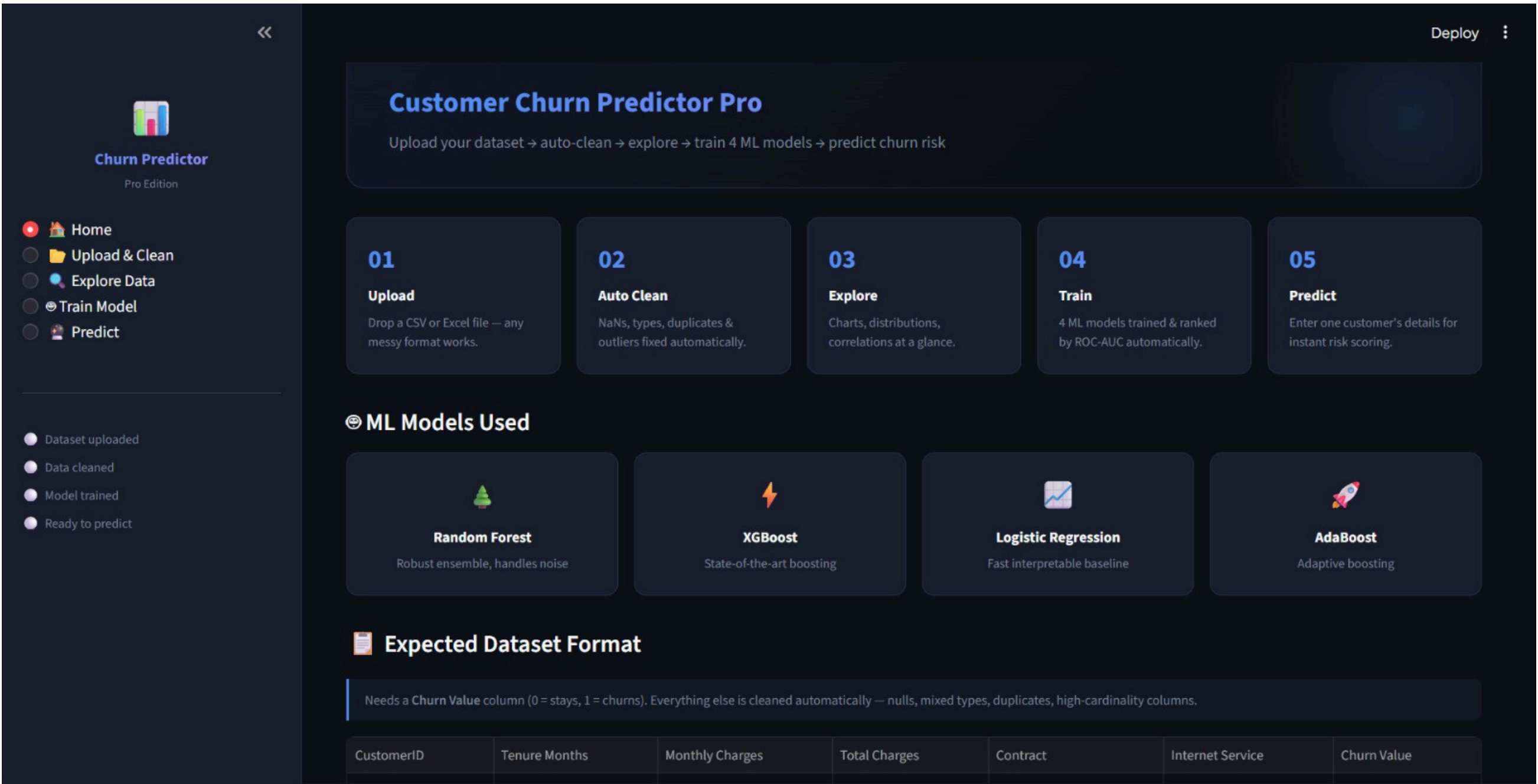
Explainability - SHAP



Deployment & Conclusion

The trained pipeline was deployed as an interactive **Streamlit web application**, enabling non-technical stakeholders to operationalize churn prediction without writing code. The app supports three core workflows:

<https://customerchurnprediction2-jur86nbrr7vjdtjsjqeqyz.streamlit.app/>



Thankyou